ECE 340: Semiconductor Devices

Lecture 11: Drift, Hall effect, Fermi level at equilibrium

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Lecture (Section C): M,W,F / 2:00-2:50pm / 3017 ECEB

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(Recap) Electron and Hole Current in E-Field

Electron Current Relationships

$$J_{x,n} = -qn\langle v_x \rangle = qn\mu_n E_x = \sigma_n E_x$$

Electron Conductivity: $\sigma_n = qn\mu_n$

Electron Mobility: $\mu_n = \frac{q\bar{t}_n}{m_n^*}$

Hole Current Relationships

$$J_{x,p} = qp\langle v_x \rangle = qp\mu_p E_x = \sigma_p E_x$$

Hole Conductivity: $\sigma_p = qp\mu_p$

Hole Mobility: $\mu_p = \frac{q\bar{t}_p}{m_p^*}$

Combined Current

$$J_{x,total} = q(n\mu_n + p\mu_p)E_x = \sigma E_x$$

$$\sigma = q(n\mu_n + p\mu_p)$$

$V = IR$, or $I = \frac{1}{R}V$

We've made it to V=IR!

(Recap) Example: Intrinsic Silicon

Assume uniform distribution of carriers: no diffusion

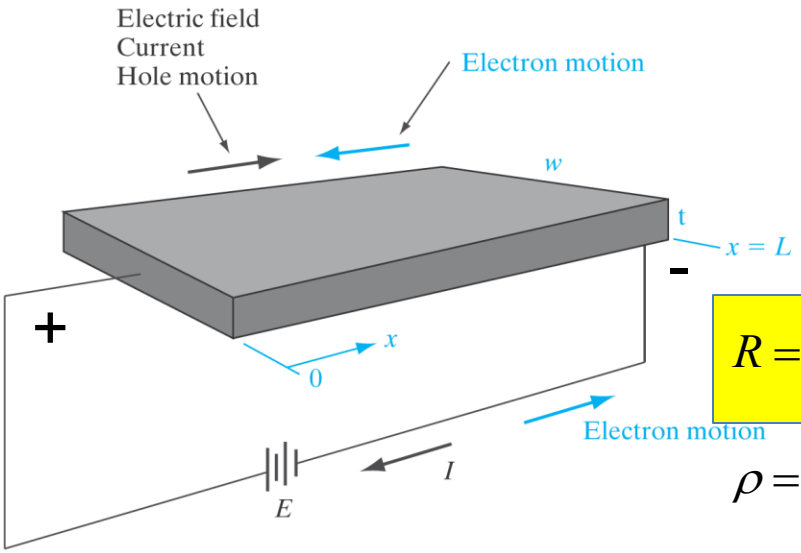


Figure 3.21

Drift of electrons and holes in a semiconductor bar.

$$J_n = -qn\langle v_n \rangle = -qn(-\mu_n E_x)$$

$$J_p = qp\langle v_p \rangle = qp(\mu_p E_x)$$

$$R = \frac{\rho L}{A} = \frac{\rho L}{wt} = \frac{L}{wt}\frac{1}{\sigma}$$

$$\rho = \frac{1}{\sigma}$$

$$\rho(\text{Intrinsic Silicon}) = \frac{1}{\sigma} = \frac{1}{q(n\mu_n + p\mu_p)}$$

$$= \frac{1}{1.6x10^{-19}(1.5\times10^{10}\,\square 1350 + 1.5\times10^{10}\,\square 480)}$$

$$= \frac{1}{4.39\times10^{-6}(\Omega - cm)^{-1}} = 2.28\times10^{5}\,\Omega - cm$$

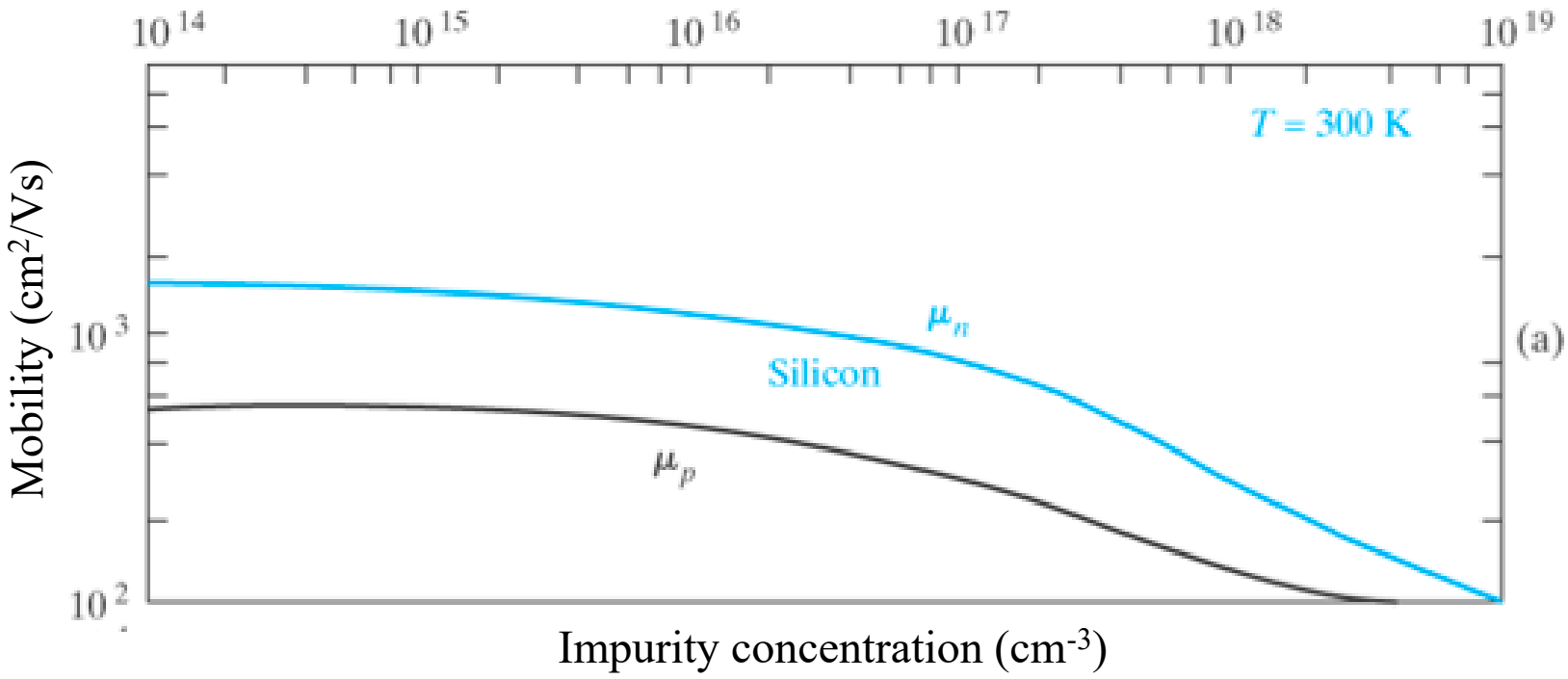
We know how to calculate n and p

What about μ?

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(Recap) What Value Do I Use for Mobility?

$$\mu \square \frac{q\bar{t}}{m^*}$$



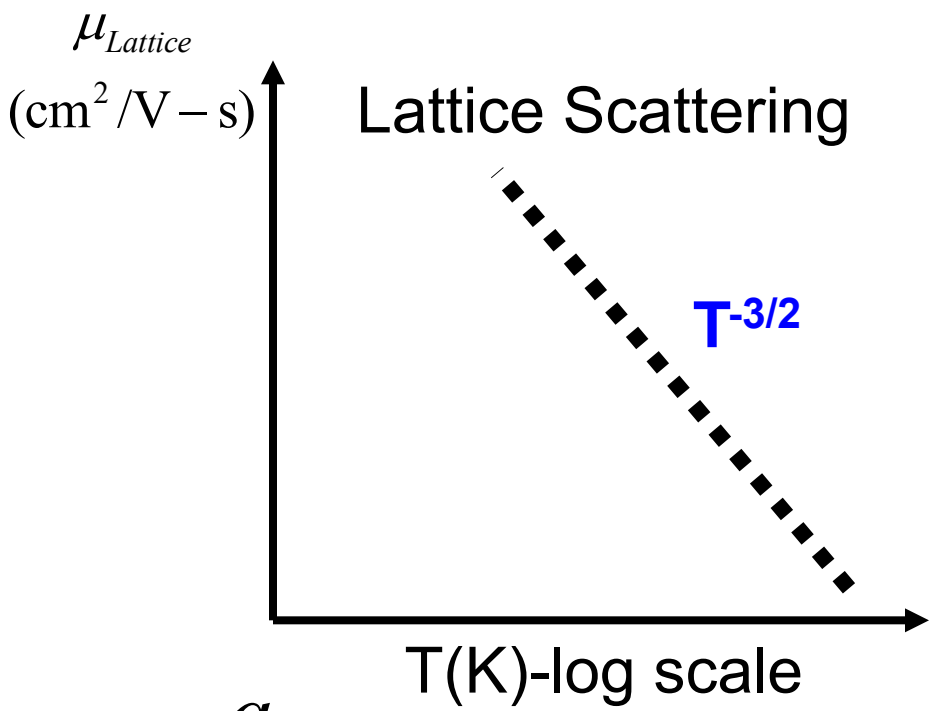
Impurity Concentration, Not Carrier Concentration

N_D+N_A

Temperature Effect on Lattice Scattering

- Electron and hole mobility is a function of the frequency of collision events (scattering)
- **Lattice** Scattering
  - A carrier moving through the crystal is scattered by a vibration of the lattice (phonon)
  - At temperature increases, there is more vibration of the lattice
  - The mobility decreases as the temperature increases due to greater thermal agitation of the lattice
  - At higher temperature, the mean time between collisions is also smaller

$$\mu_{Lattice} \propto T^{-3/2}$$

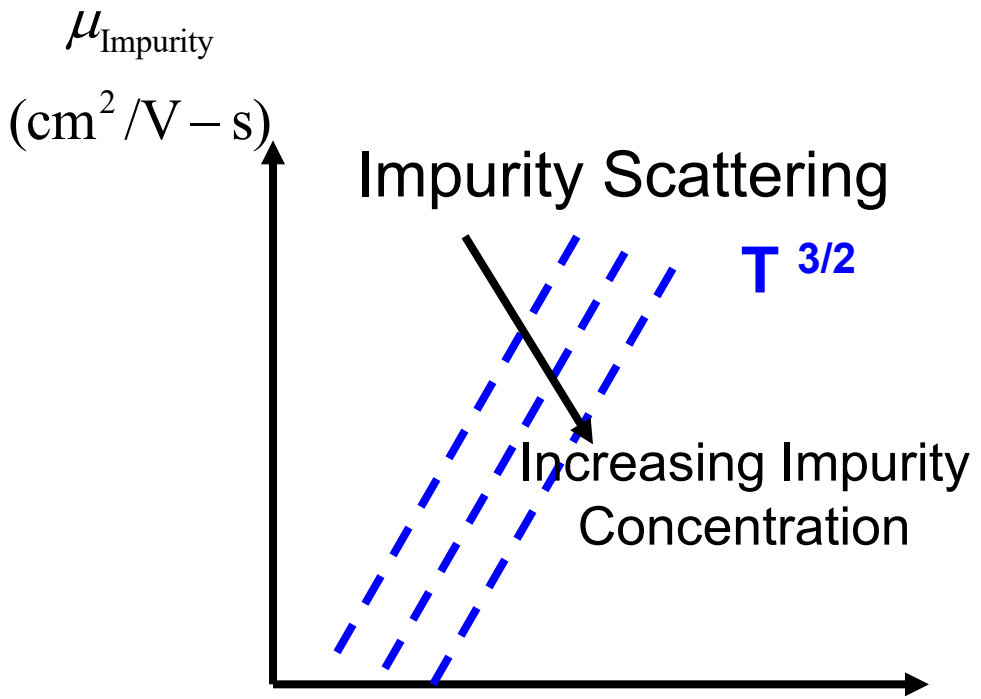


$$\mu = \overline{t}\,\frac{q}{m^*}$$

Temperature Effect on Impurity Scattering

- **Impurity** Scattering
  - Ionized impurity scattering refers to the interaction with charged ions (carrier freeze out not considered)
  - A carrier moving through the crystal is scattered by ionized impurities
  - The **mobility decreases** as the **impurity concentration increases** since there are more scattering centers
  - The **mobility decreases** as the **temperature decreases** since the carrier itself has less thermal energy and will be scattered more strongly by the impurity

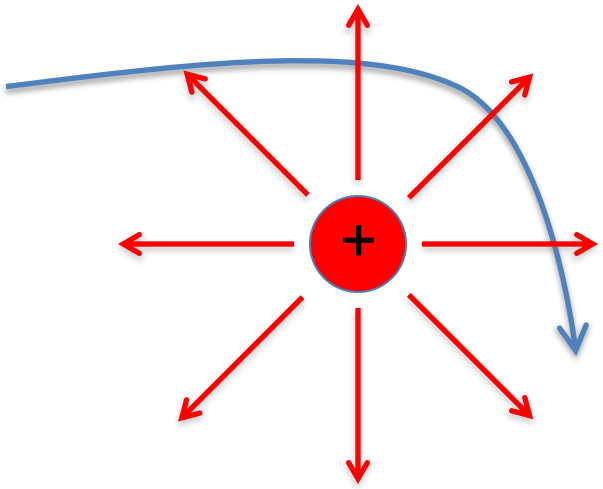
$$\mu_{\text{Impurity}} \propto T^{3/2}$$



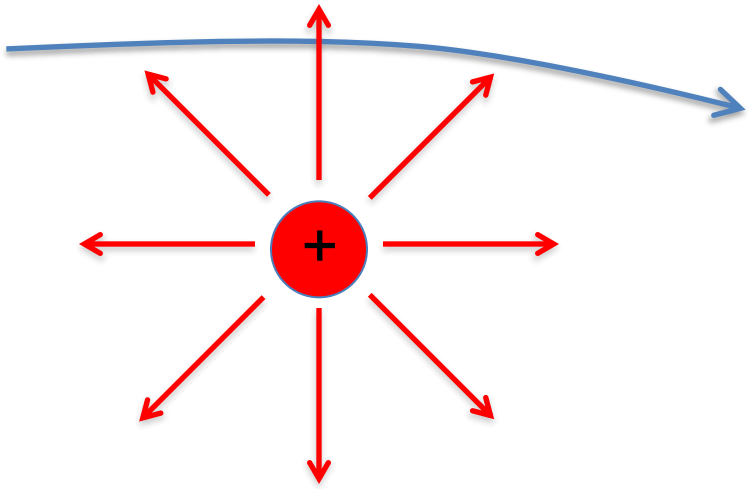
$$\mu = \overline{t}\,\frac{q}{m^*}$$

Fast Electron Versus Slow Electron

Slow Electron
(Cold Temperature)



Fast Electron
(Warm Temperature)



- The slow electron interacts with the fixed field for a longer period of time, deflecting it from the direction of propagation

Combined Effects

Mobilities Add Inversely:

$$\frac{1}{\mu_{tot}} = \frac{1}{\mu_1} + \frac{1}{\mu_2} + \frac{1}{\mu_3} + \ldots$$

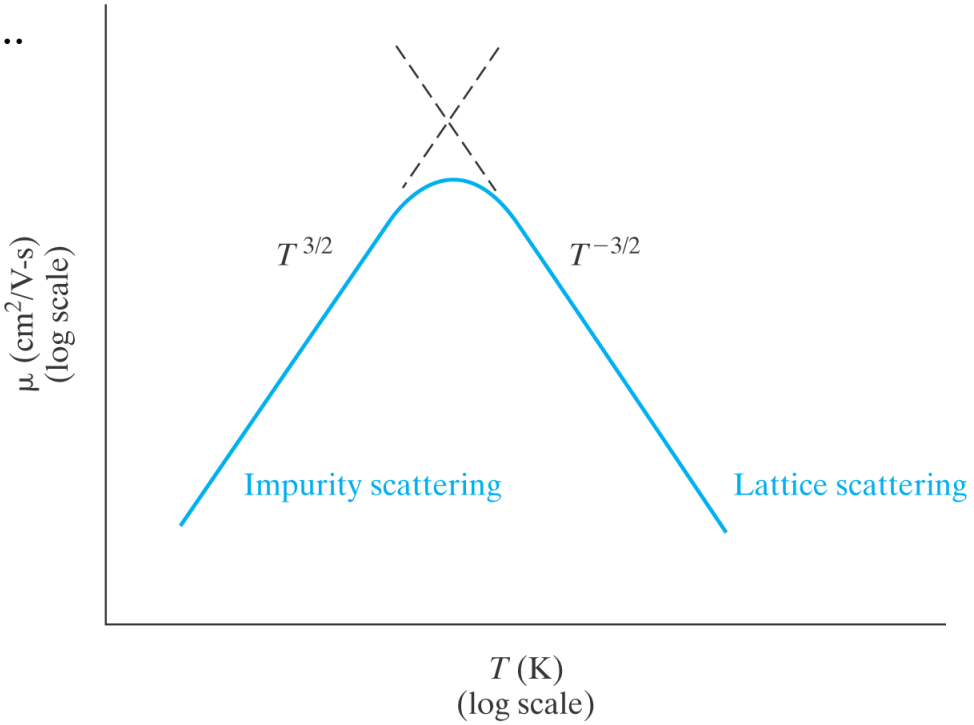


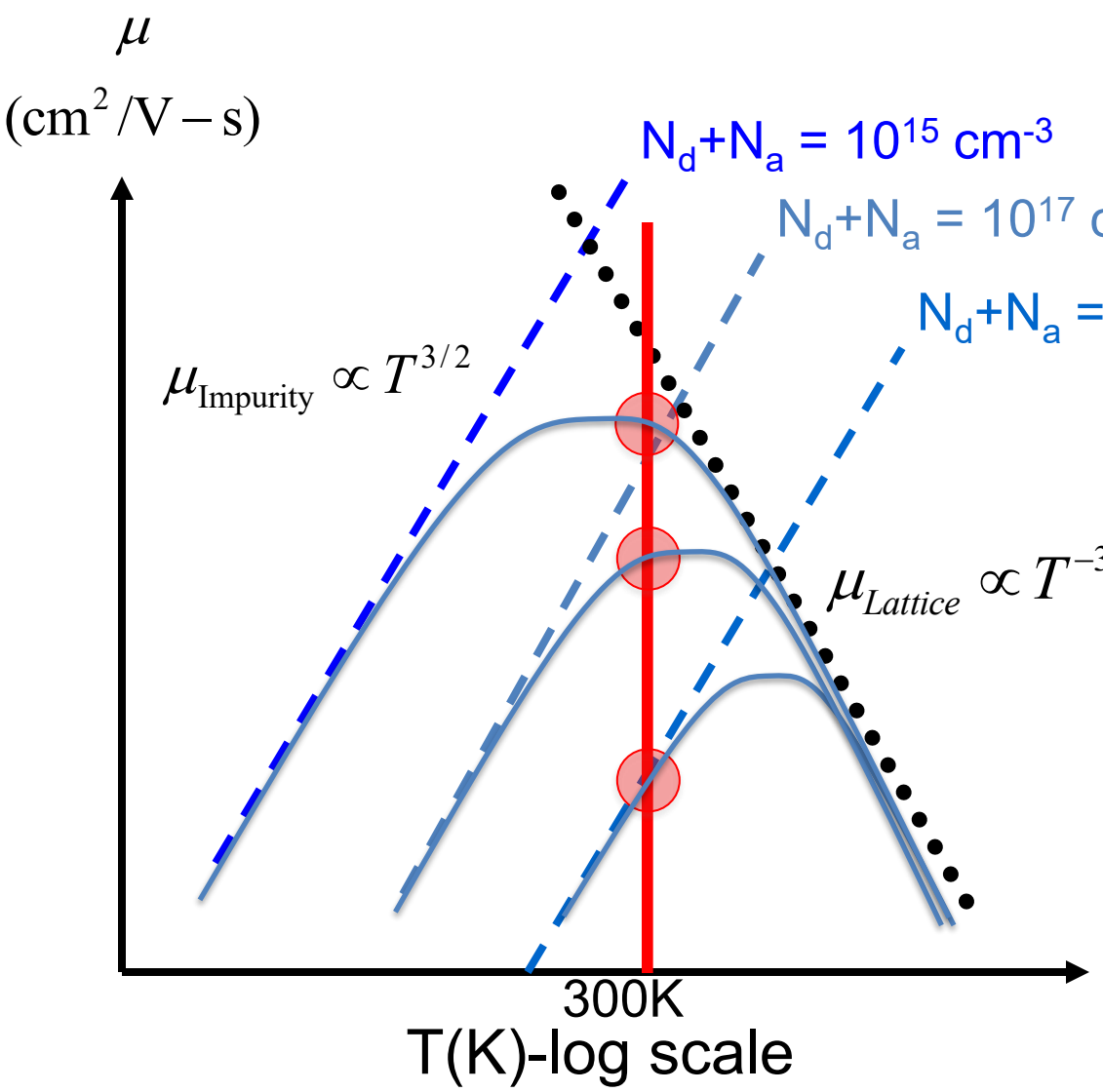
Figure 3.22

Approximate temperature dependence of mobility with both lattice and impurity scattering.

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Effects of Both Temperature and Impurity Concentration on Mobility

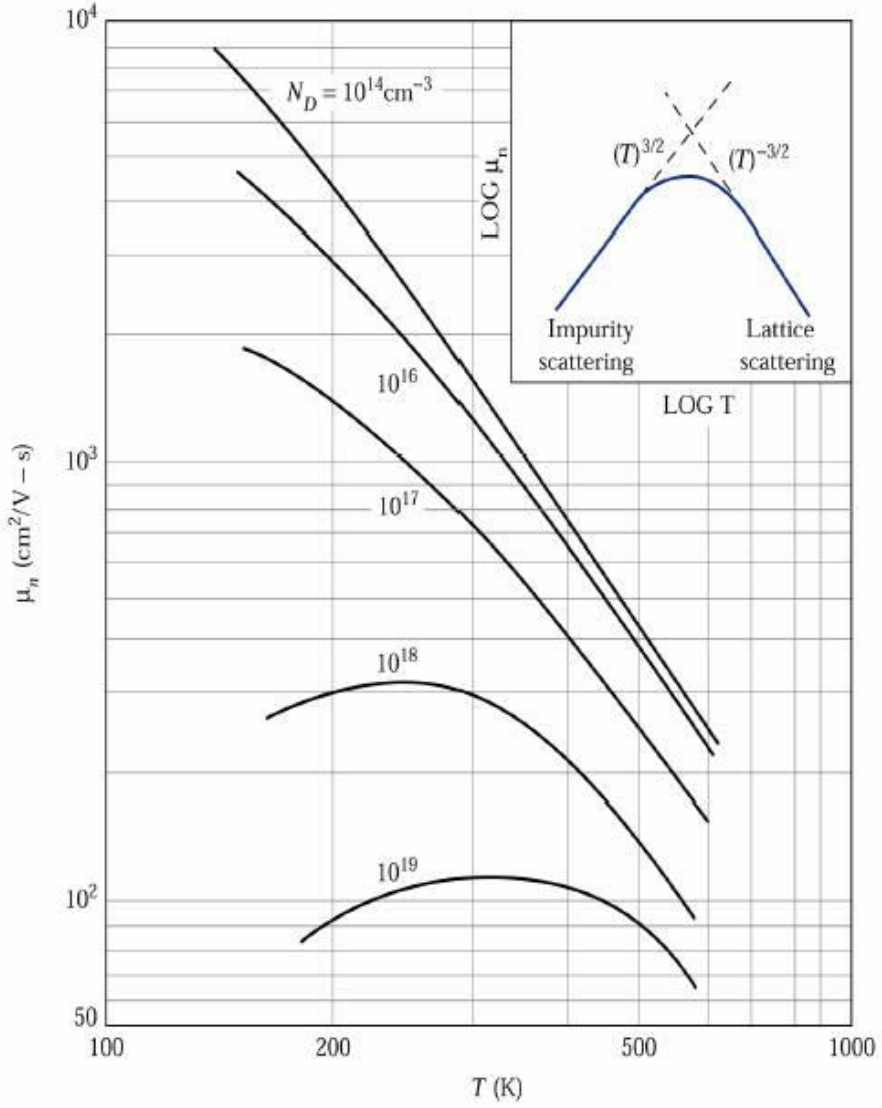
$$\frac{1}{\mu} = \frac{1}{\mu_{Impurity}} + \frac{1}{\mu_{Lattice}} + \frac{1}{\mu_{Defect}} + \ldots$$



At T=300K

- For $N_d+N_a=10^{19}$cm^{-3}, the mobility is dominated by impurity scattering
- For $N_d+N_a=10^{17}$cm^{-3}, the mobility is dominated by impurity scattering
- For $N_d+N_a=10^{15}$cm^{-3}, the mobility is dominated by lattice scattering

Comparison With Measured Data



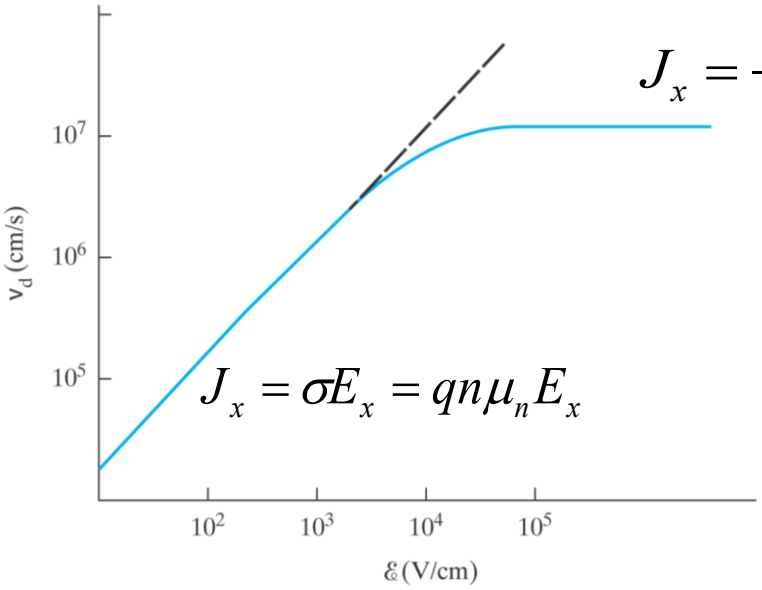
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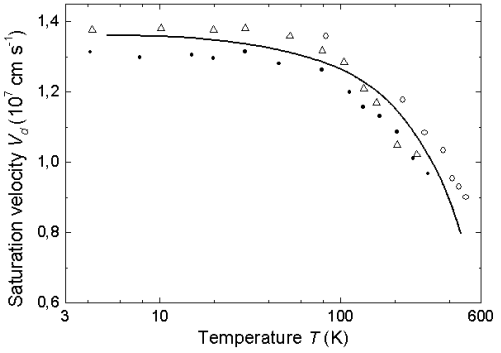
High Field Effects

Bonus Material

Saturation of Drift Velocity

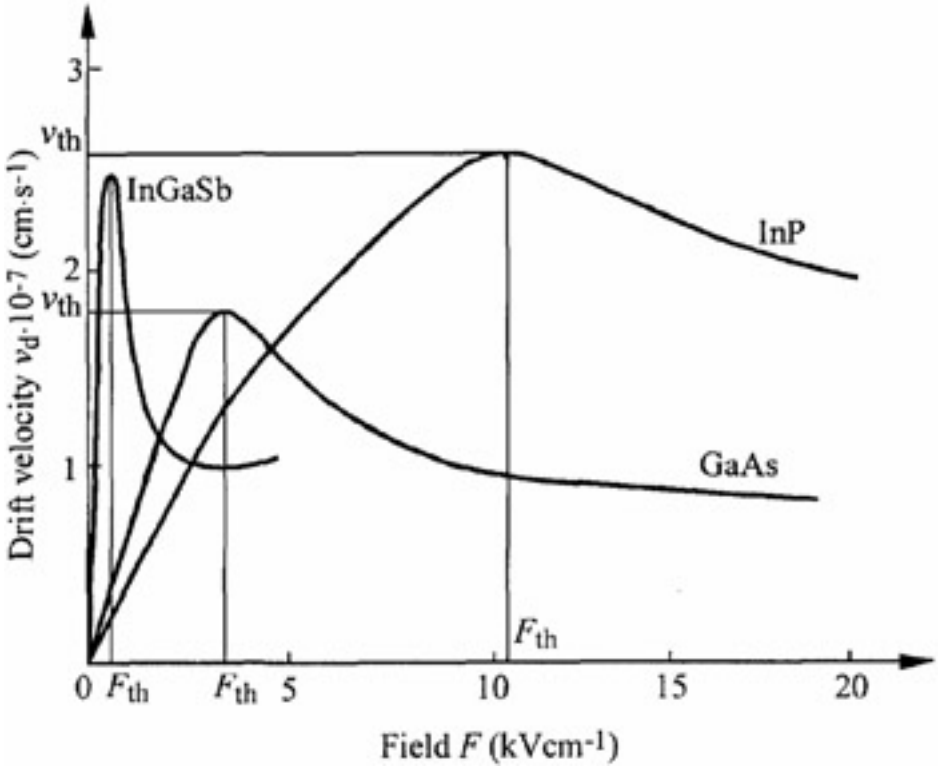
- Ohmic region: carrier drift velocity is proportional to electric field
- High field region: current saturates due to saturation of the drift velocity
  - Additional energy is transferred to the crystal lattice (phonon) rather than increasing the velocity.
  - Saturation velocity at 300K is ~10^7 cm/s in silicon
  - A "hot electron" has a velocity exceeding the mean thermal velocity in the crystal
  - Ballistic transport, or hot electrons, are used in some devices for high speed performance



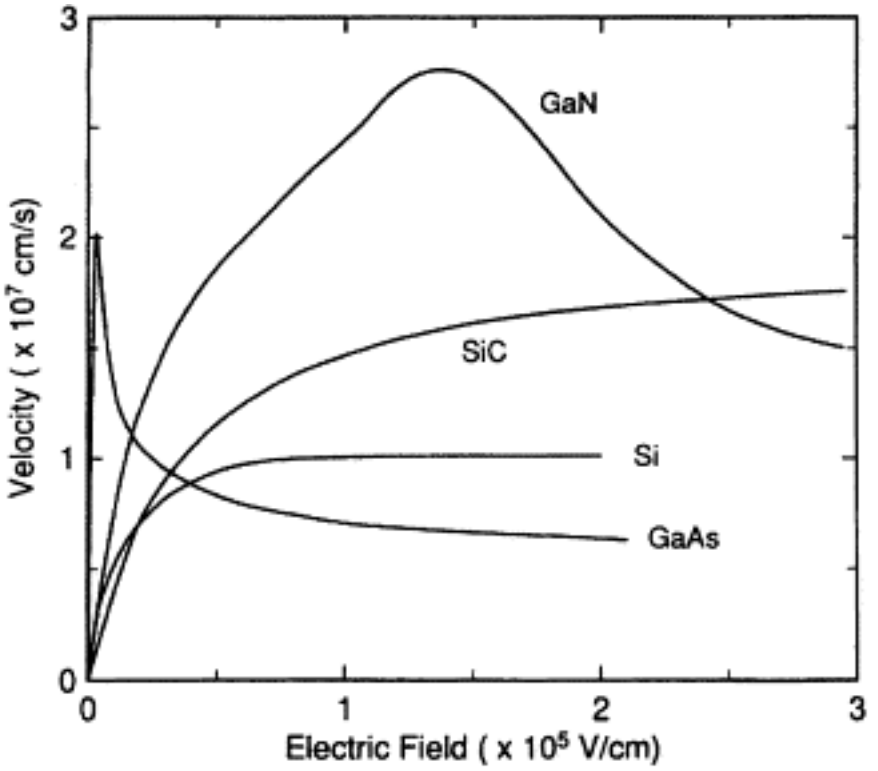


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Drift Velocity for Various Materials



http://www.globalspec.com/reference/48367/203279/4-2-avalanche-injection-in-n-n-n-p-p-p-structures



http://www.sciencedirect.com/science/article/pii/S0927796X00000280
Pearton, et. al.

A Few Cautions

- **Density of states effective mass** is used for calculating carrier concentration
  - DOS effective mass is related to 'how many states are available near the band edge'. In other words, 'how much k-space is available in all directions'

- **Conductivity effective mass** is used for calculating carrier transport (e.g., mobility), and different from the density of states effective mass for Silicon
  - Conductivity effective mass is related to 'how easily carriers move in response to an electric field'.

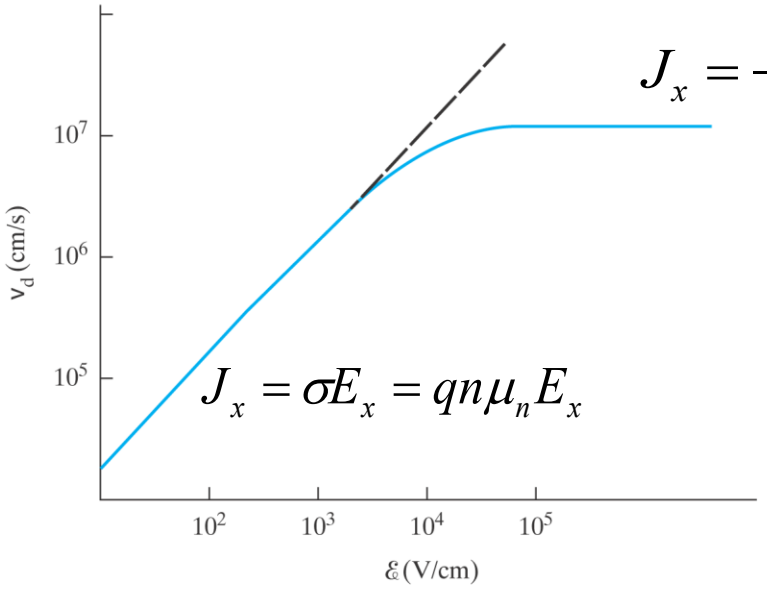
- The topic of effective mass is more complicated than can reasonably be presented in an introductory course (treatment as tensor in higher level courses)

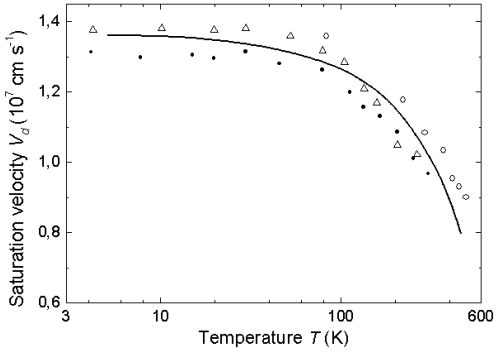
High Field Effects

Bonus Material

Saturation of Drift Velocity

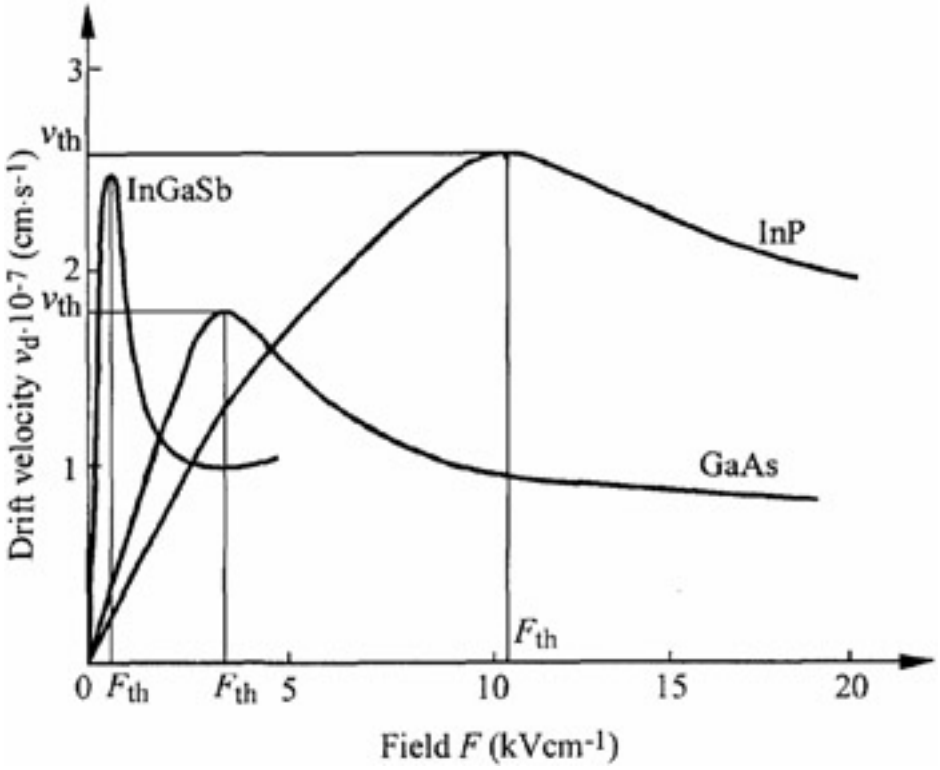
- Ohmic region: carrier drift velocity is proportional to electric field
- High field region: current saturates due to saturation of the drift velocity
  - Additional field force is directly offset by increased scattering
  - Additional energy is transferred directly to the crystal lattice
  - Mean thermal velocity at 300K is ~10^7 cm/s in silicon
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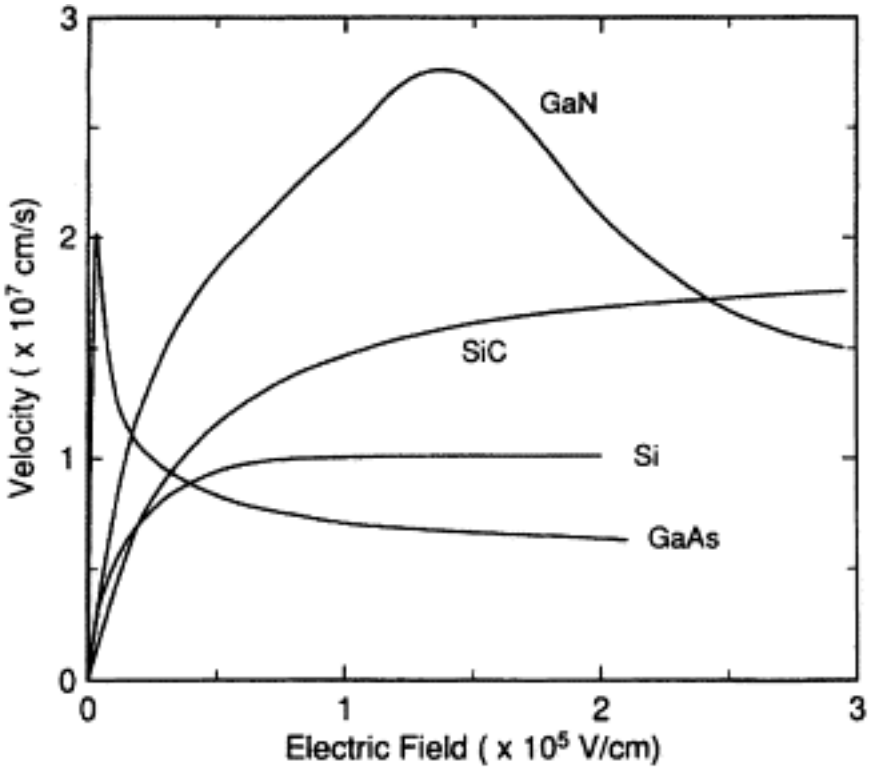


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Drift Velocity for Various Materials



http://www.globalspec.com/reference/48367/203279/4-2-avalanche-injection-in-n-n-n-p-p-p-structures



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Pearton, et. al.

L11 stopped here

The Hall Effect

Bonus Material

The Hall Effect

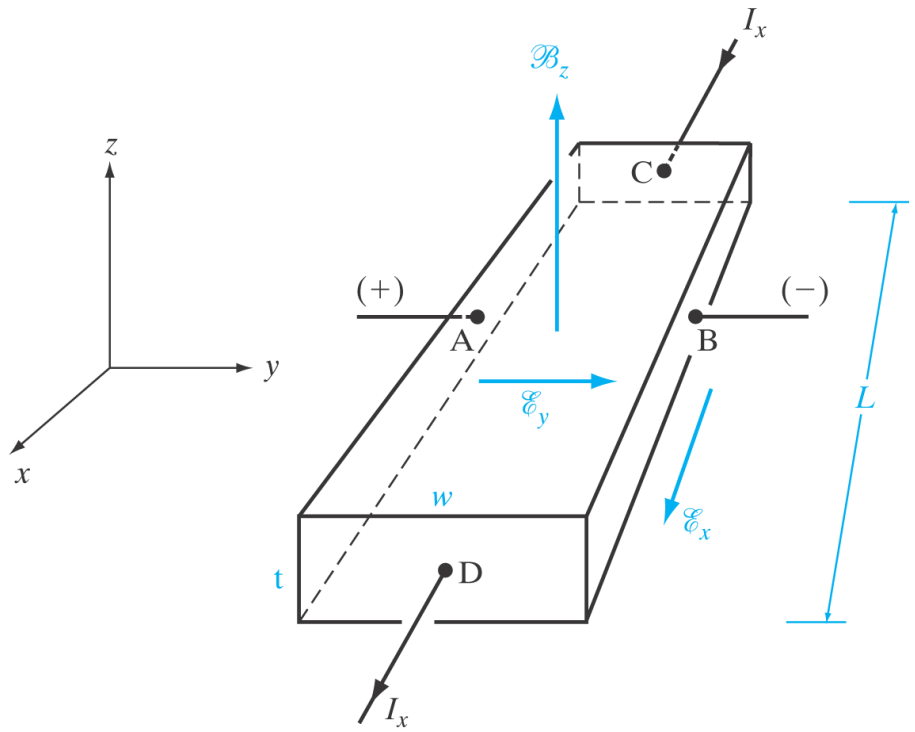


Figure 3.25

The Hall effect.

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The Hall Effect

- Used to precisely measure carrier concentration and mobility
- A magnetic field (B_z) is applied perpendicular to the direction of hole drift (x-direction) in a p-type sample.
  - Holes (and electrons) will experience a force in the –y direction (right hand rule) per Maxwell's Equations
  - Charge buildup across the sample (A face to B face) will cause a voltage to develop, called the Hall voltage
  - This field will prevent further net motion of the holes (countering force established by electric field)
- The fact that a field is established is known as the Hall Effect

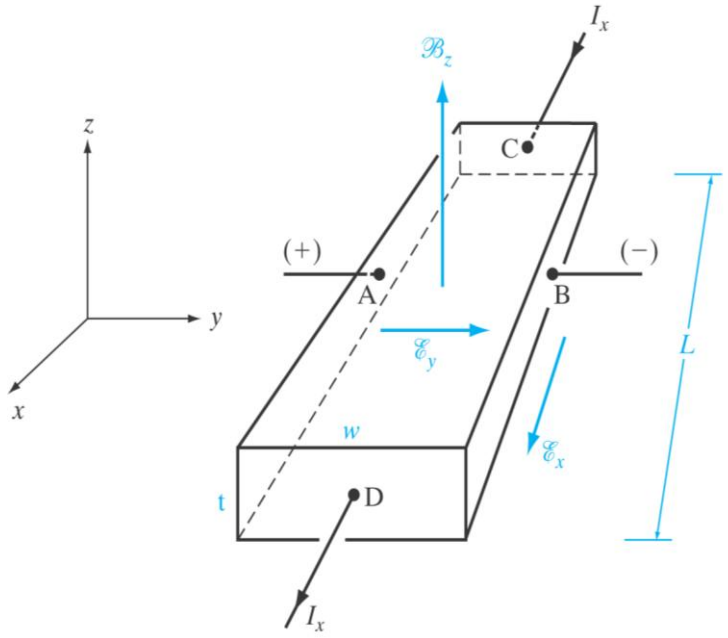


Figure 3.25

The Hall effect.

$$F = q(E + v \times B)$$

$$F_y = q(E_y - v_x B_z)$$

In Steady State: $F_y = 0$

$$E_y = v_x B_z$$

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Hall Voltage and Hall Coefficient

In Steady State, $F_y = 0$

$$E_y = v_x B_z$$

(recall, J=qpv)

$$E_y = \frac{J_x}{qp_o} B_z = R_H J_x B_z$$

Hall coefficient $R_H = \frac{1}{qp_o}$

Hall Voltage:

$$V_{AB} = E_y w$$

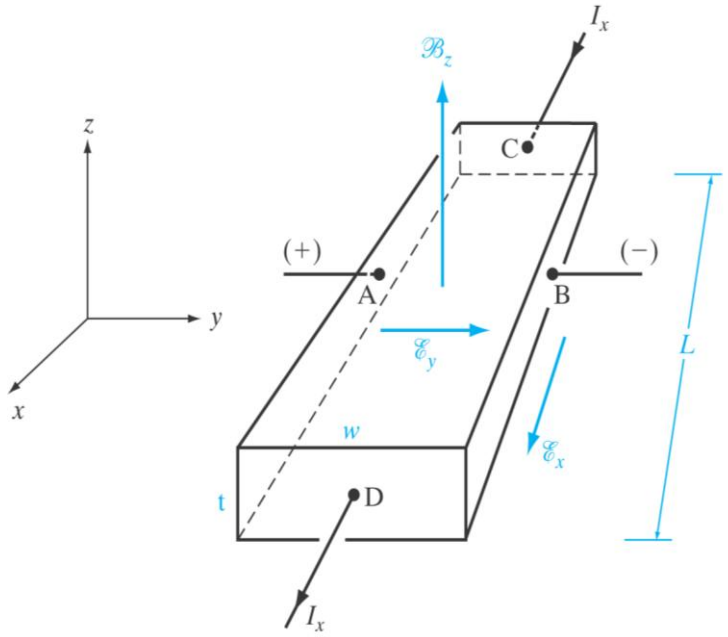


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Carrier Concentration and Mobility

Carrier Concentration

- Apply a known magnetic field B_z
- Apply a known current I_x
- Measure the Hall voltage induced across the sample
- Calculate the carrier concentration

$$E_y = v_x B_z = \frac{J_x}{qp_o} B_z$$

$$p_o = \frac{J_x}{qE_y} B_z = \frac{(I_x / wt)B_z}{q(V_{AB} / w)}$$

$$p_o = \frac{I_x B_z}{qtV_{AB}}$$

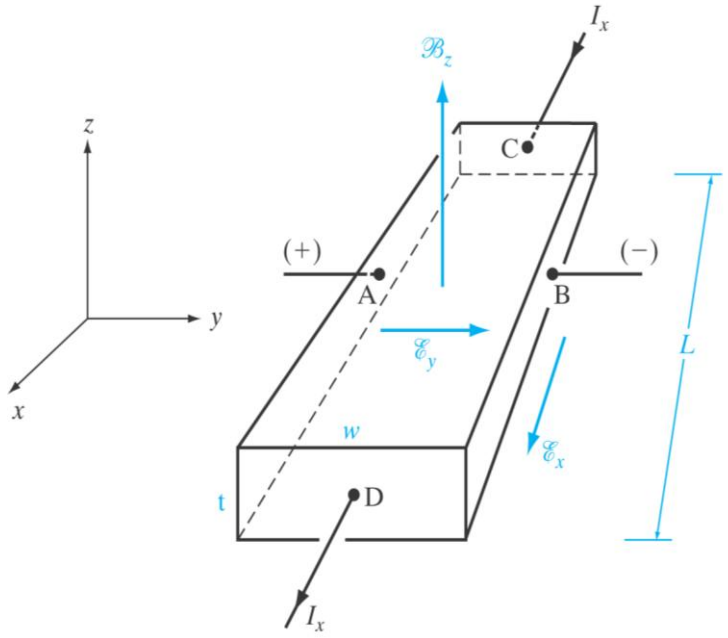


Figure 3.25

The Hall effect.

Carrier Concentration and Mobility

Mobility

- Apply a known magnetic field B_z
- Apply a known current I_x
- Measure the Hall voltage induced across the sample
- Measure the resistance across the length of the sample (V_{CD}/I_x)
- Calculate mobility

$$\rho(\Omega - cm) = R\frac{wt}{L} = \frac{V_{CD}}{I_x}\frac{wt}{L}$$

$$\sigma = \frac{1}{\rho} = q\mu_p p_o$$

$$\mu_p = \frac{1}{\rho}\frac{1}{qp_o} = \frac{R_H}{\rho}$$

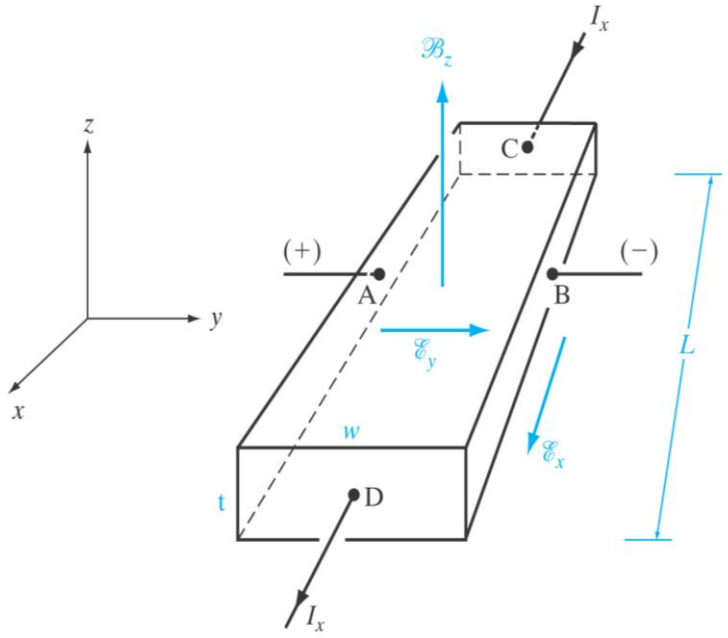
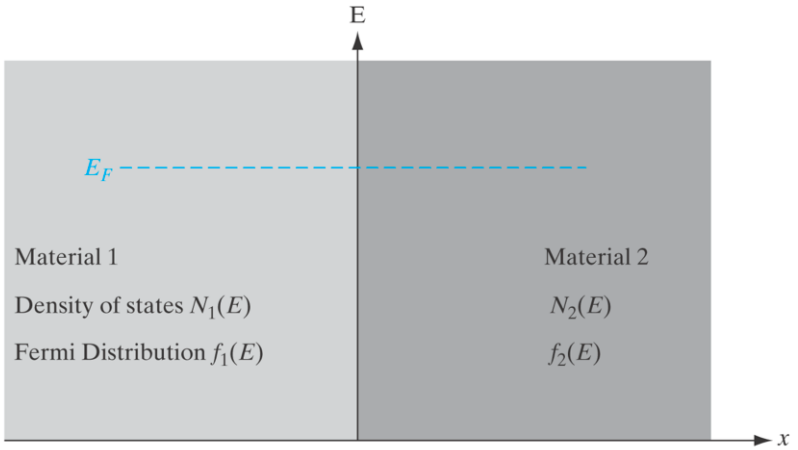


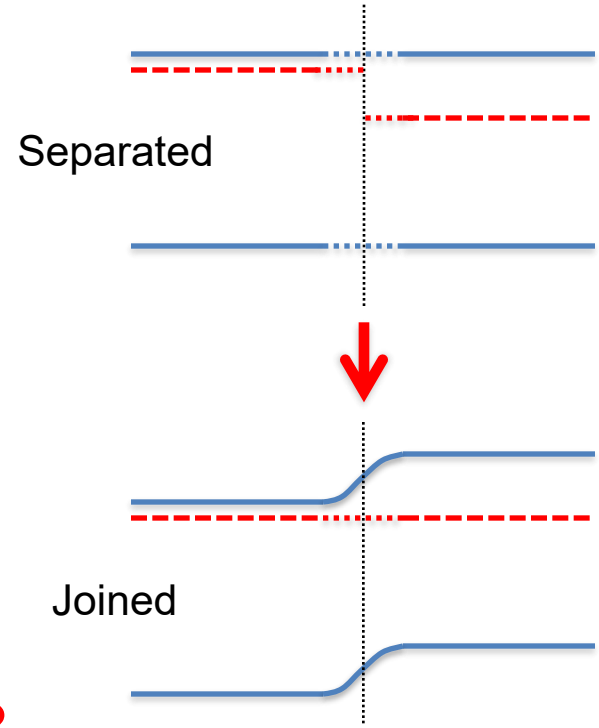
Figure 3.25
The Hall effect.

Invariance of the Fermi Level at Equilibrium

Key Concept

- No Discontinuity or Gradient Can Be Present in the **Equilibrium** Fermi Level E_F
  - In a drawing of energy versus position, E_F is a horizontal line that other energies are referenced against





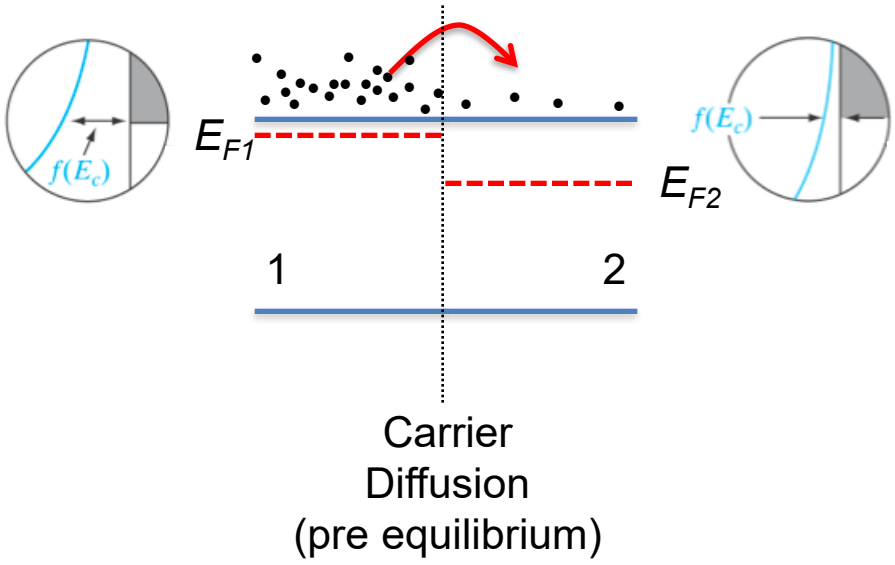
Why?

High Level Explanation

- The Fermi level tells us the probability that a state is occupied at a given energy
- Across a piece of material, a different probability of occupancy at a given energy would lead to a net motion of carriers toward the region with a lower probability
- A net motion of carriers would lead to a net current, which cannot be present in equilibrium
  - Comment: Pay attention to the word “net” – carriers can be in motion due to drift and diffusion as long as the motion is balanced. (This, in fact, is how the Fermi level equilibrates.) More on this later.

Fermi Level Misalignment: Net Flow

- "Across a piece of material, a different probability of occupancy at a given energy would lead to a net motion of carriers toward the region with a lower probability"
- A net motion of carriers is induced.



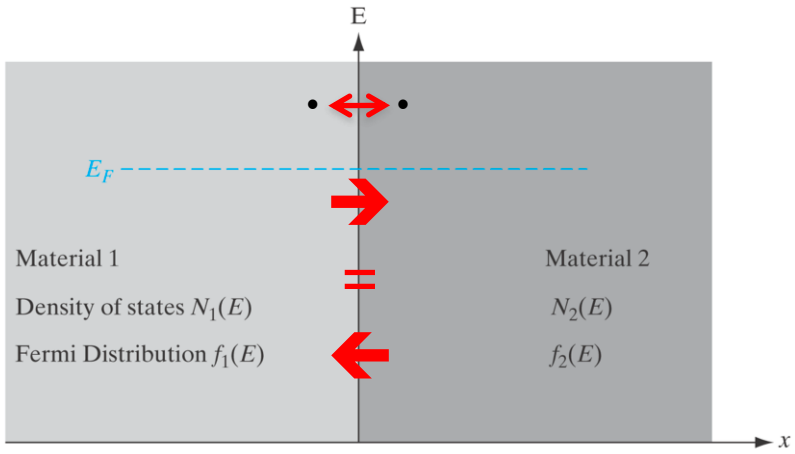
- For any given E:

  $f_1(E) > f_2(E)$ (misaligned)

Another Explanation

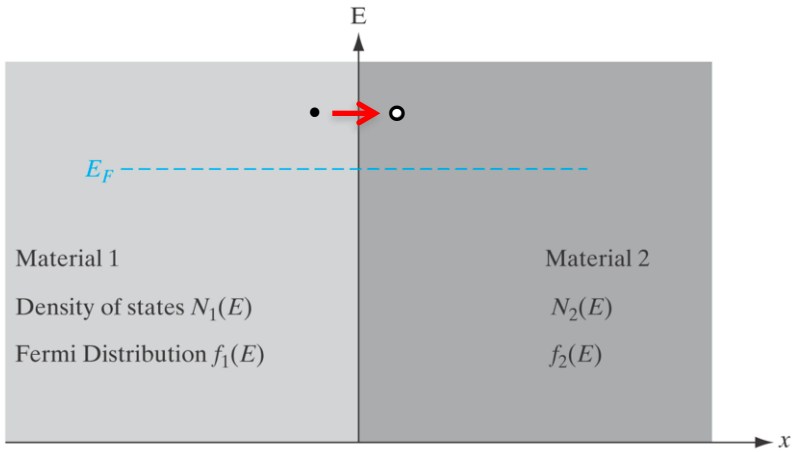
In Equilibrium, there is:

- No net current
- No net charge transport
- For 2 joined materials (1 and 2), the transfer of electrons from material 2 to 1 must exactly balance the transfer from material 1 to 2



Case 1: Electrons Moving From 1 to 2

- At a given energy "E" the motion of an electron from material 1 to material 2 requires a full state in material 1 and an empty state in material 2
- The rate of motion is proportional to the product of the number of electrons at energy "E" in material 1 and the number of empty states at energy "E" in Material 2



Density of States in Material 1: $N_1(E)$

Density of States in Material 2: $N_2(E)$

Probability State Occupied in Material 1: $f_1(E)$

Probability State Empty in Material 2: $\left[1-f_2(E)\right]$

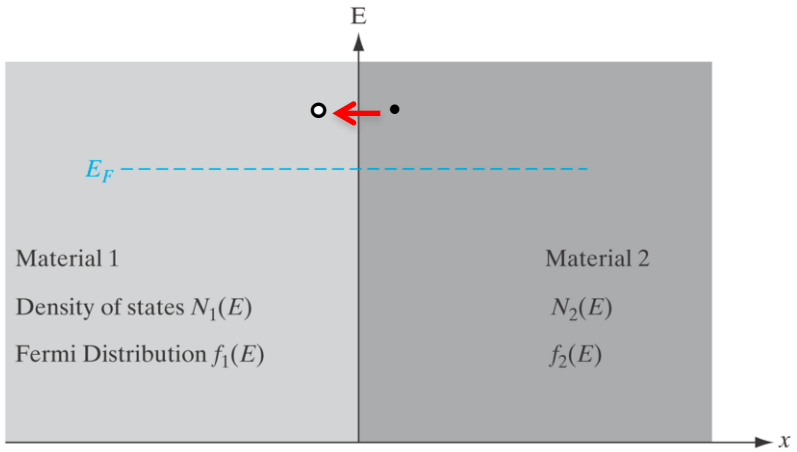
$$\text{Rate}_{1\to 2}: \quad R = a\left\{N_1(E)f_1(E)\right\}\left\{N_2(E)\left[1-f_2(E)\right]\right\}$$

electrons

empty states

Case 2: Electrons Moving From 2 to 1

- At a given energy "E" the motion of an electron from material 2 to material 1 requires a full state in material 2 and an empty state in material 1
- The rate of motion is proportional to the product of the number of electrons at energy "E" in material 2 and the number of empty states at energy "E" in Material 1



Density of States in Material 1: $N_1(E)$

Density of States in Material 2: $N_2(E)$

Probability State Empty in Material 1: $\left[1-f_1(E)\right]$

Probability State Occupied in Material 2: $f_2(E)$

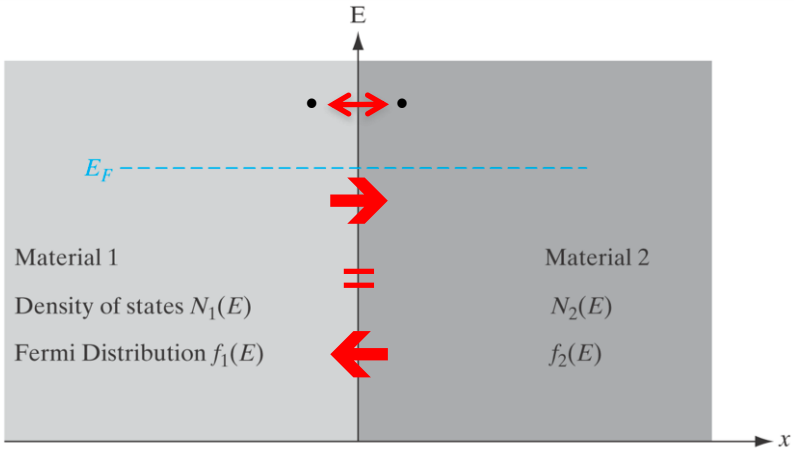
$$\text{Rate}_{2\to1}: \quad R = a\left\{N_1(E)\left[1-f_1(E)\right]\right\}\left\{N_2(E)f_2(E)\right\}$$

empty states

electrons

Equilibrium Fermi Level

- There is no net charge transfer or energy transfer between regions in equilibrium, so the rates from Case 1 and Case 2 must balance
  - We assume that the constant of proportionality "a" for the rate of motion from 1 to 2 is the same as that from 2 to 1 (reciprocity)



$$R_{1\to 2} = aN_1(E)f_1(E)\times N_2(E)\left[1-f_2(E)\right] = aN_2(E)f_2(E)\times N_1(E)\left[1-f_1(E)\right] = R_{2\to 1}$$

so

$$N_1(E)f_1(E)N_2(E) - N_1(E)f_1(E)N_2(E)f_2(E) = N_2(E)f_2(E)N_1(E) - N_2(E)f_2(E)N_1(E)f_1(E)$$

multiplying through by $\dfrac{1}{N_1(E)N_2(E)}$:

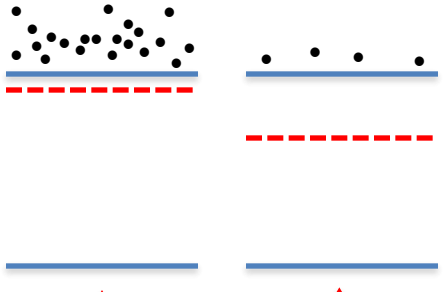
$$f_1(E) - f_1(E)f_2(E) = f_2(E) - f_2(E)f_1(E) \quad \text{therefore}$$

$$f_1(E) = f_2(E) \quad \text{and}$$

$$\frac{1}{1+e^{(E-E_{F1})/kT}} = \frac{1}{1+e^{(E-E_{F2})/kT}} \quad \Rightarrow \quad E_{F1} = E_{F2}$$

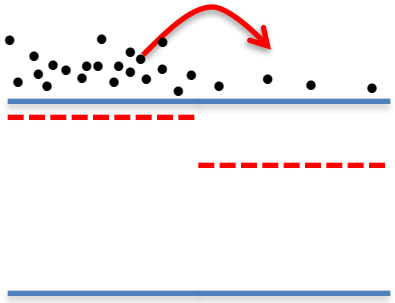
$$\boxed{\frac{dE_F}{dx} = 0}$$

Establishing Equilibrium

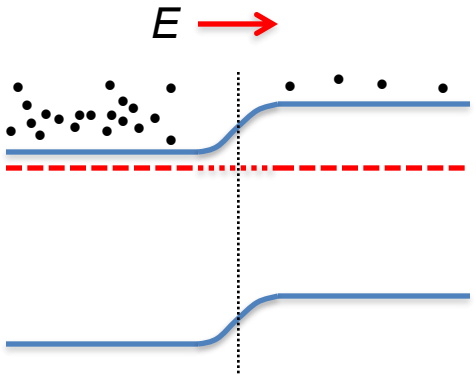


High Carrier Concentration

Low Carrier Concentration



Carrier Diffusion (pre equilibrium)



Equilibrium: Diffusion Balanced by Drift (Electric Field)

We Will Come Back to this in the Discussion of pn Junctions!